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Lab 2: Database Design and Normalization

Part 1: Entities and Attributes

Entity	Attributes	Primary key
Department	dept_id, dept_name	dept_id
Student	student_id, matric_no, full_name, dept_id	student_id
Lecturer	lecturer_id, full_name, dept_id	lecturer_id
Course	course_id, course_code, course_title	course_id
Enrolment	enrolment_id, student_id, course_id, session	enrolment_id
Teaching	teaching_id, lecturer_id, course_id, session	teaching_id

Part 1: Relationships and Cardinalities

The design has two many-to-many relationships, so it needs two junction tables: Enrolment and Teaching.

Relationship	Cardinality	Explanation
Department – Student	1:N	A student belongs to one department; a department has many students.
Department – Lecturer	1:N	A lecturer belongs to one department; a department has many lecturers.
Student – Course	M:N	A student registers for many courses; a course has many students.
Lecturer – Course	M:N	A course can have different lecturers across sessions; a lecturer can teach many courses.

Part 2: Normalization

Unnormalized Table

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

First Normal Form (1NF)

What I fixed: I removed multi-valued cells by giving each student–course pair its own row, so every cell now holds a single value.

Primary key: **(matric_no, course_code)**, a composite key, since neither column is unique on its own.

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept	session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

Second Normal Form (2NF)

What I fixed: I removed partial dependencies, columns that depended on only part of the composite key, by moving student details and course details into their own tables.

Student (PK: matric_no)

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course (PK: course_code)

course_code	course_title	lecturer	lecturer_dept
SEN307	Database Systems	Dr A. Okafor	Software Eng.
SEN301	Software Architecture	Dr M. Bello	Software Eng.
COS301	Operating Systems	Mrs F. Eze	Computer Sci.

Enrolment (PK: matric_no + course_code)

matric_no	course_code	session
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Third Normal Form (3NF)

What I fixed: I removed transitive dependencies, where a non-key column depended on another non-key column, by moving department and lecturer details into their own tables linked by foreign keys.

Department (PK: dept_id)

dept_id	dept_name
1	Software Engineering
2	Computer Science

Student (PK: student_id, FK: dept_id)

student_id	matric_no	full_name	dept_id
1	20231234	Ada Obi	1
2	20231235	John Musa	2
3	20231236	Grace Okon	1

Lecturer (PK: lecturer_id, FK: dept_id)

lecturer_id	full_name	dept_id
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Course (PK: course_id)

course_id	course_code	course_title
1	SEN307	Database Systems
2	SEN301	Software Architecture
3	COS301	Operating Systems

Enrolment (PK: enrolment_id, FKs: student_id, course_id)

enrolment_id	student_id	course_id	session
1	1	1	2026/2027
2	1	2	2026/2027
3	2	1	2026/2027
4	2	3	2026/2027
5	3	2	2026/2027

Teaching (PK: teaching_id, FKs: lecturer_id, course_id)

teaching_id	lecturer_id	course_id	session

teaching_id	lecturer_id	course_id	session
1	1	1	2026/2027
2	2	2	2026/2027
3	3	3	2026/2027

Final Schema

- Department (dept_id PK, dept_name)
- Student (student_id PK, matric_no, full_name, dept_id FK → Department)
- Lecturer (lecturer_id PK, full_name, dept_id FK → Department)
- Course (course_id PK, course_code, course_title)
- Enrolment (enrolment_id PK, student_id FK → Student, course_id FK → Course, session)
- Teaching (teaching_id PK, lecturer_id FK → Lecturer, course_id FK → Course, session)